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Communication device, non-transitory computer-readable recording medium storing computer-readable instructions for communication device, and non-transitory computer-readable recording medium storing computer-readable instructions for first external device

Abstract

A communication device may supply a first public key and specific information to a first wireless interface, send a first authentication response to the first external device in a case where a first authentication request is received from the first external device after the first public key and the specific information have been sent to the first external device. The communication device may receive first connection information from the first external device after the first authentication response has been sent to the first external device, establish the second wireless connection in a case where the first connection information is received from the first external device, and establish the third wireless connection in a case where a specific signal including the specific information is received from the first external device after the first public key and the specific information have been sent to the first external device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a Continuation of U.S. application Ser. No. 18/136,039, filed on Apr. 18, 2023, which is a Continuation of U.S. application Ser. No. 17/807,178, filed on Jun. 16, 2022, now U.S. Pat. No. 11,647,552, issued May 9, 2023, which is a Continuation of U.S. application Ser. No. 17/326,961, filed on May 21, 2021, now U.S. Pat. No. 11,399,399, issued Jul. 26, 2022, which is a Continuation of U.S. application Ser. No. 16/727,568, filed on Dec. 26, 2019, now U.S. Pat. No. 11,044,770, issued Jun. 22, 2021, which claims priority to Japanese Patent Application No. 2018-247065, filed on Dec. 28, 2018, the entire contents of which are hereby incorporated by reference into the present application.

TECHNICAL FIELD

(1) The disclosure herein discloses a technique for establishing a wireless connection between a communication device and an external device.

BACKGROUND ART

(2) DPP (Device Provisioning Protocol) scheme, which is a connection scheme developed by the Wi-Fi Alliance, is described in “Device Provisioning Protocol Specification Version 1.0”. In the DPP scheme, for example, a wireless communication according to the DPP scheme is executed in response to establishment of an NFC connection between a first device and a second device.

SUMMARY

(3) In response to the establishment of an NFC connection between the first device and the second

device, there are a situation where wireless communication according to the DPP scheme is to be executed, and a situation where wireless communication according to a connection scheme different from the DPP scheme is to be executed.

(4) The disclosure herein discloses a technique whereby, in response to the establishment of a wireless connection between a communication device and an external device, the communication device establishes a wireless connection different from the aforementioned wireless connection according to an appropriate connection scheme among a plurality of connection schemes.

(5) The disclosure herein discloses a communication device. The communication device may comprise: a first wireless interface; one or more wireless interfaces different from the first wireless interface; a processor; and a memory storing computer-readable instructions therein, the computer-readable instructions, when executed by the processor, causing the communication device to: establish a first wireless connection with a first external device via the first wireless interface; and supply a first public key that is a public key of the communication device and specific information to the first wireless interface, the first public key and the specific information being sent to the first external device by using the first wireless connection via the first wireless interface; in a case where a first authentication request in which the first public key is used is received from the first external device via a second wireless interface among the one or more wireless interfaces after the first public key and the specific information have been sent to the first external device: send a first authentication response that is a response to the first authentication request to the first external device via the second wireless interface; receive first connection information from the first external device via the second wireless interface after the first authentication response has been sent to the first external device, the first connection information being for establishing a second wireless connection between the communication device and a second external device via the second wireless interface; and in a case where the first connection information is received from the first external device, establish, according to a first connection scheme, the second wireless connection between the communication device and the second external device via the second wireless interface by using the first connection information; and in a case where a specific signal including the specific information is received from the first external device via a third wireless interface among the one or more wireless interfaces after the first public key and the specific information have been sent to the first external device, establish, according to a second connection scheme different from the first connection scheme, a third wireless connection between the communication device and the first external device via the third wireless interface.

(6) A computer program for implementing the above communication device, and a computer-readable medium storing the computer program are also novel and useful. Moreover, a method carried out by the communication device is also novel and useful.

(7) The disclosure further discloses a non-transitory computer-readable recording medium storing computer-readable instructions for a first external device. The first external device may comprise a first wireless interface and one or more wireless interfaces different from the first wireless interface, and the computer-readable instructions, when executed by a processor of the first external device, may cause the first external device to: establish a first wireless connection with a communication device via the first wireless interface; receive a public key of the communication device and specific information from the communication device by using the first wireless connection via the first wireless interface; in a case where the public key and the specific information are received from the communication device, determine which of a second wireless connection and a third wireless connection is to be established; in a case where it is determined that the second wireless connection is to be established: send an authentication request in which the public key is used to the communication device via a second wireless interface among the one or more wireless interfaces; in a case where the authentication request is sent to the communication device, receive an authentication response that is a response to the authentication request from the communication device via the second wireless interface; and in a case where the authentication

response is received from the communication device, send connection information to the communication device via the second wireless interface, the connection information being for establishing the second wireless connection according to a first connection scheme between the communication device and the second external device, wherein in the communication device, the second wireless connection is established between the communication device and the second external device by using the connection information in a case where the connection information is received from the first external device; and in a case where it is determined that the third wireless connection is to be established: send a specific signal including the specific information to the communication device via a third wireless interface among the one or more wireless interfaces; and establish, according to a second connection scheme different from the first connection scheme, the third wireless connection via the third wireless interface between the communication device and the first external device after the specific signal has been sent to the communication device.

(8) A computer-readable medium storing the above computer program is also novel and useful. Further, the first external device itself, and a method carried out by the first external device are also novel and useful.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a configuration of a communication system.
- (2) FIG. 2 shows an explanatory diagram for explaining an outline of Case A in which a Wi-Fi connection according to DPP scheme is established between a printer and an access point.
- (3) FIG. 3 shows a sequence view of a process of Bootstrapping.
- (4) FIG. 4 shows a sequence view of a process of Authentication.
- (5) FIG. 5 shows a sequence view of a process of Configuration.
- (6) FIG. 6 shows a sequence view of a process of Network Access.
- (7) FIG. 7 shows a sequence view of a process of Case B in which a WFD connection is established between a terminal and the printer.
- (8) FIG. 8 shows a sequence view of a process of Case C in which the terminal and the printer have established a Wi-Fi connection with an access point.
- (9) FIG. 9 shows a sequence view of a process of Case D in which a mode shifting operation is executed in the printer.
- (10) FIG. 10 shows a flowchart of a process executed by the terminal.
- (11) FIG. 11 shows a flowchart of a process executed by the printer.
- (12) FIG. 12 shows a flowchart of a process executed by a terminal of a second embodiment.
- (13) FIG. 13 shows a flowchart of a process executed by a printer of the second embodiment.

EMBODIMENTS

- (14) (Configuration of Communication System 2; FIG. 1)
- (15) As shown in FIG. 1, a communication system 2 comprises an access point (hereinbelow termed simply as “AP”) 6, a terminal 10, and a printer 100. In this embodiment, a situation is assumed in which a user uses the terminal 10 to establish a wireless connection according to the Wi-Fi scheme (hereinbelow termed as “Wi-Fi connection”) between the printer 100 and the AP 6.
- (16) (Configuration of Terminal 10)
- (17) The terminal 10 is a portable terminal device such as a cellphone (e.g., a smartphone), a PDA, and a tablet PC. In a variant, the terminal 10 may be a desktop PC, a laptop PC or the like. The terminal 10 comprises an operation unit 12, a display unit 14, a Wi-Fi interface 16, an NFC interface 18, and a controller 30. The units 12 to 30 are connected to a bus line (reference number omitted). Hereinbelow, interface is termed simply as “I/F”.
- (18) The operation unit 12 comprises a plurality of keys. The user can input various instructions to

the terminal **10** by operating the operation unit **12**. The display unit **14** is a display for displaying various information.

(19) The Wi-Fi I/F **16** is a wireless interface configured to execute a Wi-Fi communication according to the Wi-Fi scheme. The Wi-Fi scheme is a wireless communication scheme for executing wireless communication according to, for example, 802.11 standard of IEEE (The Institute of Electrical and Electronics Engineers, Inc.) and standards complying thereto (such as 802.11a, 11b, 11g, 11n, etc.). The Wi-Fi I/F **16** can establish a Wi-Fi connection with the AP **6** in a case where an SSID (Service Set Identifier) for identifying a wireless network in which the AP **6** operates as a parent station, and a password used in that wireless network are input via the operation unit **12**. Further, even if this information is not input via the operation unit **12**, the Wi-Fi I/F **16** can establish a Wi-Fi connection with the AP **6** by using a so-called PIN code scheme or PBC (Push Button Configuration) scheme. Hereinbelow, a Wi-Fi connection established in this manner may be termed “normal Wi-Fi connection”.

(20) Further, the Wi-Fi I/F **16** supports the WFD (Wi-Fi Direct (registered trademark)) scheme that has been formulated by the Wi-Fi Alliance, and the DPP (Device Provisioning Protocol) scheme that has been formulated by the Wi-Fi Alliance. The WFD scheme is a connection scheme described in the standard “Wi-Fi Peer-to-Peer (P2P) Technical Specification Version 1.1” created by the Wi-Fi Alliance. In the WFD standard, three states are defined as states of a WFD device: Group Owner state (hereinbelow termed “G/O state”), client state, and device state. The WFD device is capable of selectively operating in one of the above three states. The DPP scheme is described in the standard “Device Provisioning Protocol Specification Version 1.0” created by the Wi-Fi Alliance, and is a connection scheme for easily establishing a Wi-Fi connection between a pair of devices (such as the printer **100** and the AP **6**) by using the terminal **10**. Hereinbelow, a Wi-Fi connection established according to the DPP scheme may be termed “DPP connection”, and a Wi-Fi connection established according to the WFD scheme may be termed “WFD connection”.

(21) The NFC I/F **18** is an I/F for executing NFC communication according to the NFC scheme. The NFC scheme is a wireless communication scheme based on, e.g. international standards ISO/IEC 14443, 15693, 18092, etc. I/Fs called NFC Forum Devices, and I/Fs called NFC Forum Tags are known as types of I/F for executing NFC communication. In this embodiment, the NFC I/F **18** is an NFC Forum Device.

(22) Here, differences between the Wi-Fi I/F and the NFC I/F will be described. A communication speed of a wireless communication via a Wi-Fi I/F (e.g., maximum communication speed is 11 to 600 Mbps) is faster than a communication speed of a wireless communication via an NFC I/F (e.g., maximum communication speed is 100 to 424 Kbps). Further, a frequency used in wireless communication via a Wi-Fi I/F (e.g. a 2.4 GHz band or 5.0 GHz band) is different from a frequency used in wireless communication via an NFC I/F (e.g. a 13.56 MHz band). Further, a maximum distance at which wireless communication via a Wi-Fi I/F can be executed (e.g., about 100 m at maximum) is greater than a maximum distance at which wireless communication via an NFC I/F can be executed (e.g., about 10 cm at maximum).

(23) The controller **30** comprises a CPU **32** and a memory **34**. The CPU **32** executes various processes in accordance with an OS (Operating System) program **36** stored in the memory **34**. The memory **34** is configured by a volatile memory, a non-volatile memory, or the like. Further, the memory **34** stores a print application **38** (hereinbelow simply termed “app **38**”). The app **38** is a program for causing the printer **100** to execute printing, and is installed on the terminal **10** from, for example, a server on the Internet provided by a vendor of the printer **100**.

(24) (Configuration of Printer **100**)

(25) The printer **100** is a peripheral device (e.g., a peripheral device of the terminal **10**, etc.) capable of executing a print function. The printer **100** comprises an operation unit **112**, a display unit **114**, a Wi-Fi I/F **116**, an NFC I/F **118**, a print executing unit **120**, and a controller **130**. The units **112** to **130** are connected to a bus line (reference number omitted).

(26) The operation unit **112** comprises a plurality of keys. The user can input various instructions to the printer **100** by operating the operation unit **112**. The display unit **114** is a display for displaying various information. The print executing unit **120** comprises printing mechanism such as an ink jet scheme, laser scheme, etc.

(27) The Wi-Fi I/F **116** is identical to the Wi-Fi I/F **16** of the terminal **10**. That is, the Wi-Fi I/F **116** supports the WFD scheme and the DPP scheme. For this reason, the printer **100** can establish a Wi-Fi connection with the AP **6** according to the DPP scheme (that is, a DPP connection), and can further establish a Wi-Fi connection with the terminal **10** according to the WFD scheme (that is, a WFD connection). The printer **100** can also establish a normal Wi-Fi connection with the AP **6**.

(28) The controller **130** comprises a CPU **132** and a memory **134**. The CPU **132** executes various processes in accordance with a program **136** stored in the memory **134**. The memory **134** is configured by a volatile memory, a non-volatile memory, or the like.

(29) The memory **134** further stores a WFD flag **138**. The WFD flag **138** indicates either a value “ON”, meaning that the printer **100** is in a state of being capable of operating according to the WFD scheme, or “OFF”, meaning that the printer **100** is in a state of being incapable of operating according to the WFD scheme. The state in which the WFD flag **138** is “ON” means that the operation state of the printer **100** is one of the three states (that is, G/O state, client state, and device state) of the WFD scheme. The state in which the WFD flag **138** is “OFF” means that the operation state of the printer **100** is not any of the three states of the WFD scheme.

Specific Examples; FIG. 2 to FIG. 9

(30) Next, specific examples of processes executed by each of the devices **6**, **10**, **100** will be described with reference to FIGS. 2 to 9. Firstly, a process of Case A in which a DPP connection between the printer **100** and the AP **6** is established will be described with reference to FIGS. 2 to 6.

Overview of Case A; FIG. 2

(31) Firstly, an overview of the Case A will be described with reference to FIG. 2. As aforementioned, the terminal **10** and the printer **100** support the DPP scheme, and further the AP **6** also supports the DPP scheme. In this embodiment, the DPP connection between the printer **100** and the AP **6** is established by each of the devices **6**, **10**, **100** executing communication according to the DPP scheme. Hereinbelow, to facilitate understanding, operations which CPUs (such as the CPUs **32**, **132**) of the respective devices execute will be described with the devices (such as the terminal **10**, the printer **100**) as subjects of action instead of describing the operations with the CPUs as the subjects of action.

(32) In T5, the terminal **10** executes Bootstrapping (hereinbelow termed simply as “BS”) according to the DPP scheme with the AP **6**. This BS is a process of providing information that is to be used in Authentication (hereinbelow termed simply as “Auth”) of T10 (to be described later) from the AP **6** to the terminal **10** in response to a QR code (registered trademark) attached to the AP **6** being captured by the terminal **10**.

(33) In T10, the terminal **10** executes Auth according to the DPP scheme with the AP **6** by using the information obtained in the BS of T5. This Auth is a process for each of the terminal **10** and the AP **6** to authenticate their communication counterpart.

(34) In T15, the terminal **10** executes Configuration (hereinbelow termed simply as “Config”) according to the DPP scheme with the AP **6**. This Config is a process of sending information for establishing the DPP connection between the printer **100** and the AP **6** to the AP **6**. Specifically, in the Config, the terminal **10** creates a Configuration Object (hereinbelow, Configuration Object is simply be termed “CO”) for AP for establishing the DPP connection between the printer **100** and the AP **6**, and sends the CO for AP to the AP **6**. As a result, the CO for AP is stored in the AP **6**.

(35) Next, the terminal **10** executes BS according to the DPP scheme with the printer **100** in T20. This BS is a process of providing information that is to be used in Auth of T25 (to be described later) from the printer **100** to the terminal **10** by using the NFC connection established between the

NFC I/F **18** of the terminal **10** and the NFC I/F **118** of the printer **100**.

(36) In **T25**, the terminal **10** executes Auth according to the DPP scheme with the printer **100** by using the information obtained in the BS of **T20**. This Auth is a process for each of the terminal **10** and the printer **100** to authenticate their communication counterpart.

(37) In **T30**, the terminal **10** executes Config according to the DPP scheme with the printer **100**. This Config is a process of sending information for establishing the DPP connection between the printer **100** and the AP **6** to the printer **100**. In this Config, the terminal **10** creates a CO for printer for establishing the DPP connection between the printer **100** and the AP **6**, and sends the CO for printer to the printer **100**. As a result, the CO for printer is stored in the printer **100**.

(38) In **T35**, the printer **100** and the AP **6** use the stored Cos for AP and for printer to execute Network Access (hereinbelow termed simply as “NA”) according to the DPP scheme. The NA is a process of sharing a connection key for establishing the DPP connection between the printer **100** and the AP **6**.

(39) In **T40**, the printer **100** and the AP **6** execute 4 way-handshake communication. In at least a part of the 4 way-handshake communication, the printer **100** and the AP **6** communicate encrypted information encrypted by the connection key that has been already shared in the NA in **T35**. Further, in a case where decryption of the encrypted information succeeds, the DPP connection is established between the printer **100** and the AP **6**. Due to this, the printer **100** can participate, as a child station, in a wireless network formed by the AP **6**, as a result of which the printer **100** can execute communication via the AP **6** with other device(s) participating in the wireless network. In a variant, the printer **100** and the AP **6** may execute Simultaneous Authentication of Equals (SAE, also called “Dragonfly”) communication, instead of the 4 way-handshake communication.

(40) In **T45**, the printer **100** causes the display unit **114** to display a completion screen indicating that the DPP connection has been established with the AP **6**. When the process of **T45** is completed, the process of FIG. 2 is terminated.

(41) In the DPP scheme, in order to establish the Wi-Fi connection between the printer **100** and the AP **6**, the user does not need to input information of the wireless network in which the AP **6** operates as a parent station (such as an SSID and a password) to the printer **100**. As such, the user can easily establish the Wi-Fi connection between the printer **100** and the AP **6**.

(42) (Description on Respective Processes; FIGS. 3 to 6)

(43) Next, details of the respective processes executed in **T20** to **T35** of FIG. 2 will be described with reference to FIGS. 3 to 9. Since the processes of **T5** to **T15** are similar to the processes of **T20** to **T30** except that the AP **6** is used instead of the printer **100**, the detailed description thereof will be omitted.

(44) (Bootstrapping (BS); FIG. 3)

(45) First, the process of BS of **T20** of FIG. 2 will be described with reference to FIG. 3. In an initial state of FIG. 3, the memory **134** of the printer **100** stores the public key PPK1 and the secret key psk1 of the printer **100** in advance. Further, in the initial state of FIG. 3, the WFD flag **138** of the printer **100** indicates “OFF”.

(46) In response to accepting a power ON operation from the user in **T100**, the printer **100** determines in **T102** that a normal Wi-Fi connection with the AP **6** has not been established, and shifts an operation mode of the printer **100** from non-setting mode to setting mode. The printer **100** determines that a normal Wi-Fi connection with the AP **6** has not been established in a case where AP information (for example, SSID, password, etc.) for establishing the normal Wi-Fi connection with the AP **6** is not stored in the memory **134**, or in a case where a confirmation signal for confirming whether a normal Wi-Fi connection has been established is sent via the Wi-Fi I/F **116** and a response to this confirmation signal is not received from the AP **6**. The non-setting mode is a mode in which BS information used for establishing a DPP connection between the printer **100** and the AP **6** is not supplied to the NFC I/F **118** in response to the establishment of the NFC connection. The setting mode is a mode in which the BS information is supplied to the NFC I/F **118**

in response to the establishment of the NFC connection. In other words, the non-setting mode is a mode in which a DPP connection scheme cannot be established according to the DPP, and the setting mode is a mode in which a DPP connection can be established according to the DPP scheme.

(47) In response to the app **38** of the terminal **10** being activated, and the terminal **10** being brought close to the printer **100** by the user in **T103**, an NFC connection is established between the NFC I/F **18** of the terminal **10** and the NFC I/F **118** of the printer **100** in **T104**. In this case, in **T105** the printer **100** obtains a signal indicating that the NFC connection has been established from the NFC I/F **118**, determines that the operation mode of the printer **100** is the setting mode, and supplies WFD information and the BS information to the NFC I/F **118**. The WFD information is information used for establishing a WFD connection between the terminal **10** and the printer **100**, and includes an SSID “wfd”. The SSID “wfd” is information for identifying wireless network in which the printer **100** operating in the G/O state operates as a parent station. The SSID “wfd” may be stored in advance in the memory **134**, or may be created by the printer **100** in response to the establishment of the NFC connection. The BS information includes the public key PPK1 of the printer **100** stored in the memory **134**, a channel list stored in advance in the memory **134**, and a MAC address of the printer **100**. The channel list is information indicating a plurality of communication channels determined in advance in the printer **100** (that is, a plurality of communication channels that can be used by the printer **100**).

(48) In **T106**, the printer **100** changes the WFD flag **138** in the memory **134** from “OFF” to “ON”, and shifts from the device state to the G/O state.

(49) In **T108**, the printer **100** shifts from a non-respondent state to a respondent state. The non-respondent state is a state in which the Wi-Fi I/F **116** does not send a DPP Authentication Response (hereinbelow simply termed “ARes”) (see **T210** of FIG. 4 to be described later) even when a DPP Authentication Request (hereinbelow simply termed “AReq”) is received from the terminal **10** (see **T220** to be described later). The respondent state is a state in which the Wi-Fi I/F **116** sends the ARes to the terminal **10** in response to receiving the AReq from the terminal **10**. That is, the printer **100** shifts to a state of being able to execute the Auth (see **T25** of FIG. 2) by shifting from the non-respondent state to the respondent state. Specifically, in this embodiment, the non-respondent state is a state in which even when the Wi-Fi I/F **116** receives a signal from outside, the Wi-Fi I/F **116** does not supply the signal to the CPU **132**. Further, the respondent state is a state in which in response to receiving a signal from outside, the Wi-Fi I/F **116** supplies the signal to the CPU **132** and sends a response to this signal. Since the respondent state is a state in which the CPU **132** processes the signal received from outside, processing load in that state is higher than that in the non-respondent state. In a variant, the non-respondent state may be a state in which electricity is not supplied to the Wi-Fi I/F **116**, and the respondent state may be a state in which electricity is supplied to the Wi-Fi I/F **116**. Further, in another variant, the non-respondent state may be a state in which even when the Wi-Fi I/F **116** receives the AReq from outside, the Wi-Fi I/F **116** does not supply a notification that the AReq has been received to the CPU **132**, and the respondent state may be a state in which in response to receiving the AReq from outside, the Wi-Fi I/F **116** supplies the notification that the AReq has been received to the CPU **132**.

(50) In **T110**, the NFC I/F **118** of the printer **100** sends the WFD information and the BS information to the terminal **10** by using the established NFC connection.

(51) When the app **38** is activated in **T103**, the terminal **10** executes following processes in accordance with the app **38**. In a case where the terminal **10** receives the WFD information and the BS information from the printer **100** via the NFC I/F **18** in **T110**, the terminal **10** causes the display unit **14** to display a selection screen in **T112**. The selection screen includes an “AP” button indicating that a process for establishing a DPP connection between the printer **100** and the AP **6** is to be executed, and a “terminal” button indicating that a process for establishing a WFD connection between the terminal **10** and the printer **100** is to be executed. That is, in other words, the selection

screen is a screen for causing the user to select one connection scheme from among the DPP scheme and the WFD scheme.

(52) In **T120**, the terminal **10** accepts from the user an operation for selecting the “AP” button in the selection screen. When the process of **T120** is completed, the process of **FIG. 3** is terminated.

(53) (Authentication (Auth); **FIG. 4**)

(54) Next, the process of the Auth in **T25** of **FIG. 2** will be described with reference to **FIG. 4**. In response to the “AP” button in the selection screen being selected by the user in **T120** of **FIG. 3**, the terminal **10** creates a public key **TPK1** and a secret key **tsk1** of the terminal **10** in **T200**. Next, in **T202**, the terminal **10** creates a shared key **SK1** according to Elliptic curve Diffie-Hellman key exchange (ECDH) by using the created secret key **tsk1** and the public key **PPK1** of the printer **100** received in **T110** of **FIG. 3**. Then, in **T204**, the terminal **10** creates encrypted data **ED1** by using the created shared key **SK1** to encrypt a random value **RV1**.

(55) In **T210**, the terminal **10** sends an AReq via the Wi-Fi I/F **16** to the printer **100** by setting the MAC address of the printer **100** received in **T110** of **FIG. 3** as its destination. The AReq is a signal for requesting the printer **100** to execute authentication. Here, the terminal **10** repeats sending the AReq to the printer **100** by sequentially using the plurality of communication channels in the channel list received in **T110**. The AReq includes the public key **TPK1** of the terminal **10** created in **T200**, the encrypted data **ED1** created in **T204**, and a capability of the terminal **10**.

(56) The capability is information that is pre-designated in a device supporting the DPP scheme, and includes any one of following values: a value indicating that this device is capable of operating only as a Configurator according to the DPP scheme, a value indicating that this device is capable of operating only as an Enrollee according to the DPP scheme, and a value indicating that this device is capable of operating as whichever one of the Configurator and the Enrollee. The Configurator refers to a device configured to send a CO used in the NA (**T35** of **FIG. 2**) to an Enrollee in the Config (**T30** of **FIG. 2**). On the other hand, the Enrollee refers to a device that receives the CO used in the NA from the Configurator in the Config. As above, in this embodiment, the terminal **10** creates the COs for AP and for printer and sends them respectively to the AP **6** and the printer **100**. As such, the capability of the terminal **10** includes the value indicating that it is capable of operating only as the Configurator.

(57) The printer **100** receives the AReq from the terminal **10** via the Wi-Fi I/F **116** in **T210**. As above, this AReq is sent with the MAC address of the printer **100** as the destination. As such, the printer **100** can suitably receive this AReq from the terminal **10**.

(58) Further, when the printer **100** shifts to the respondent state in **T108** of **FIG. 3**, it monitors receipt of the AReq using one communication channel among the plurality of communication channels in the channel list. As above, the AReq in **T210** is sent by sequentially using the plurality of communication channels in the channel list. As such, the printer **100** can suitably receive this AReq from the terminal **10**.

(59) Next, the printer **100** executes following processes for authenticating the sender of the AReq (that is, the terminal **10**). Specifically, firstly, in **T212**, the printer **100** creates a shared key **SK1** according to the ECDH by using the public key **TPK1** of the terminal **10** in the AReq and the secret key **psk1** of the printer **100** stored in advance in the memory **134**. Here, the shared key **SK1** created by the terminal **10** in **T202** and the shared key **SK1** created by the printer **100** in **T212** are identical to each other. Thus, the printer **100** can suitably decrypt the encrypted data **ED1** in the AReq by using the created shared key **SK1** in **T214**, as a result of which it can obtain the random value **RV1**. In a case where the decryption of the encrypted data **ED1** succeeds, the printer **100** determines that the sender of the AReq is the device with which the NFC connection has been established in **T104** of **FIG. 3**, that is, determines that the authentication succeeded, and executes subsequent processes from **T216**. On the other hand, in a case where the decryption of the encrypted data **ED1** does not succeed, the printer **100** determines that the sender of the AReq is not the device with which the NFC connection has been established in **T104**, that is, determines that the authentication failed, and

does not execute the subsequent processes from T216.

(60) In T216, the printer **100** creates a new public key PPK2 and a new secret key psk2 of the printer **100**. In a variant, the public key PPK2 and the secret key psk2 may be stored in advance in the memory **134**. Next, in T217, the printer **100** creates a shared key SK2 according to the ECDH by using the public key TPK1 of the terminal **10** in the AReq of T210 and the created secret key psk2 of the printer **100**. Then, in T218, the printer **100** creates encrypted data ED2 by using the created shared key SK2 to encrypt the obtained random value RV1 and a new random value RV2.

(61) In T220, the printer **100** sends an ARes to the terminal **10** via the Wi-Fi I/F **116**. This ARes includes the public key PPK2 of the printer **100** created in T216, the encrypted data ED2 created in T218, and a capability of the printer **100**. This capability includes the value indicating that the printer **100** is capable of operating only as the Enrollee.

(62) In response to receiving the ARes from the printer **100** via the Wi-Fi I/F **16** in T220, the terminal **10** executes following processes for authenticating the sender of the ARes (that is, the printer **100**). Specifically, firstly in T222, the terminal **10** creates a shared key SK2 according to the ECDH by using the secret key tsk1 of the terminal **10** created in T200 and the public key PPK2 of the printer **100** in the ARes. Here, the shared key SK2 created by the printer **100** in T217 and the shared key SK2 created by the terminal **10** in T222 are identical to each other. Thus, the terminal **10** can suitably decrypt the encrypted data ED2 in the ARes by using the created shared key SK2 in T224, as a result of which it can obtain the random values RV1 and RV2. In a case where the decryption of the encrypted data ED2 succeeds, the terminal **10** determines that the sender of the ARes is the device with which the NFC connection has been established in T104 of FIG. 3, that is, determines that the authentication succeeded, and executes subsequent processes from T230. On the other hand, in a case where the decryption of the encrypted data ED2 does not succeed, the terminal **10** determines that the sender of the ARes is not the device with which the NFC connection has been established in T104, that is, determines that the authentication failed, and does not execute the subsequent processes from T230.

(63) In T230, the terminal **10** sends a Confirm to the printer **100** via the Wi-Fi I/F **16**. The Confirm includes information indicating that the terminal **10** operates as the Configurator and the printer **100** operates as the Enrollee. As a result, the terminal **10** determines to operate as the Configurator in T232, and the printer **100** determines to operate as the Enrollee in T234. When the process of T234 is completed, the process of FIG. 4 is terminated.

(64) (Configuration (Config); FIG. 5)

(65) Next, the process of Config in T30 of FIG. 2 will be described with reference to FIG. 5. In T300, the printer **100** sends a DPP Configuration Request (hereinbelow termed simply as “CReq”) to the terminal **10** via the Wi-Fi I/F **116**. This CReq is a signal requesting the CO (that is, the information for establishing the DPP connection between the printer **100** and the AP **6**) to be sent.

(66) The terminal **10** receives the CReq from the printer **100** in T300 via the Wi-Fi I/F **16**. In this case, the terminal **10** obtains a group ID “Group”, a public key TPK2, and a secret key tsk2 from a memory **34** of the terminal **10** in T301. As aforementioned, the terminal **10** have already executed the Config in T15 of FIG. 2 with the AP **6**, and at that occasion the terminal **10** creates the group ID “Group”, the public key TPK2, and the secret key tsk2 and stored the same in the memory. The group ID “Group” is information for identifying a wireless network formed by the DPP connection being established between the printer **100** and the AP **6**. In a variant, a character string designated by the user may be used as the group ID. That is, in T301, the terminal **10** obtains the respective information that were stored in T15 of FIG. 2. Next, in T302, the terminal **10** creates the CO for printer (see T30 of FIG. 2). Specifically, the terminal **10** executes following processes.

(67) The terminal **10** creates a hash value HV by hashing the public key TPK2 of the terminal **10**. Further, the terminal **10** creates a specific value by hashing a combination of the hash value HV, the group ID “Group”, and the public key PPK2 of the printer **100** in the ARes in T220 of FIG. 4. Then, the terminal **10** creates a digital signature DSpr by using the secret key tsk2 of the terminal

10 to encrypt the created specific value in accordance with an Elliptic Curve Digital Signature Algorithm (ECDSA). As a result, the terminal **10** can create a Signed-Connector for printer (hereinbelow, the Signed-Connector is termed simply as “SCont”) including the hash value HV, the group ID “Group”, the public key PPK2 of the printer **100**, and the digital signature DSpr. Further, the terminal **10** creates the CO for printer including the SCont for printer and the public key TPK2 of the terminal **10**.

(68) In T310, the terminal **10** sends a DPP Configuration Response (hereinbelow termed simply as “CRes”) including the CO for printer to the printer **100** via the Wi-Fi I/F **16**.

(69) The printer **100** receives the CRes from the terminal **10** in T310 via the Wi-Fi I/F **116**. In this case, the printer **100** stores the CO for printer in the CRes in the memory **134** in T312. When the process of T312 is completed, the process of FIG. 5 is terminated.

(70) (Network Access (NA); FIG. 6)

(71) Next, the process of the NA in T35 of FIG. 2 executed between the printer **100** and the AP **6** will be described with reference to FIG. 6. As aforementioned, the processes of T5 to T15 of FIG. 2 have already been executed between the terminal **10** and the AP **6**, similarly to T20 to T30 of FIG. 2. However, the AP **6** does not execute the processes of T102 to T110 of FIG. 3. The AP **6** stores in advance a public key APK1 and a secret key ask1 of the AP **6**. Further, a QR code, which is obtained by coding the public key APK1 of the AP **6**, a channel list of the AP **6**, and a MAC address of the AP **6**, is adhered to a housing of the AP **6**. Processes similar to the processes from T200 of FIG. 4 onward are executed between the terminal **10** and the AP **6** when the terminal **10** captures this QR code. As a result, the AP **6** stores a public key APK2 and a secret key ask2 of the AP **6** (see T216 of FIG. 4), and further stores the CO for AP received from the terminal **10** (see T312 of FIG. 5). The CO for AP includes a SCont for AP and a public key TPK2 of the terminal **10**. This public key TPK2 is identical to the public key TPK2 included in the CO for printer. Further, the SCont for AP includes a hash value HV, a group ID “Group”, the public key APK2 of the AP **6**, and a digital signature DSap. This hash value HV and this group ID “Group” are respectively identical to the hash value HV and the group ID “Group1” included in the CO for printer. The digital signature DSap is information in which a specific value, which is obtained by hashing a combination of the hash value HV, the group ID “Group”, and the public key APK2, is encrypted by the secret key tsK2 of the terminal **10**, and is a value different from the digital signature DSpr included in the CO for printer.

(72) In T400, the printer **100** sends a DPP Peer Discovery Request (hereinbelow termed simply as “DReq”) including the SCont for printer to the AP **6** via the Wi-Fi I/F **116**. This DReq is a signal requesting the AP **6** to execute authentication and send the SCont for AP.

(73) In response to receiving the DReq from the printer **100** in T400, the AP **6** executes a process for authenticating the sender of the DReq (that is, the printer **100**) and the respective information in the DReq (that is, the hash value HV, the “Group”, and the public key PPK2). Specifically, in T402, the AP **6** firstly executes a first AP determination process that is regarding whether or not the hash value HV and the group ID “Group” in the received SCont for printer are respectively identical to the hash value HV and the group ID “Group” in the SCont for AP included in the stored CO for AP. In the case of FIG. 6, the AP **6** determines “identical” in the first AP determination process, thus it determines that the authentication of the sender of the DReq (that is, the printer **100**) succeeds. Here, the fact that the hash value HV in the received SCont for printer is identical to the hash value HV in the SCont for AP included in the stored CO for AP means that the SCont for printer and the SCont for AP were created by the same device (that is, the terminal **10**). As such, the AP **6** also determines that authentication of the creator of the received SCont for printer (that is, the terminal **10**) succeeds. Further, the AP **6** decrypts the digital signature DSpr in the received SCont for printer by using the public key TPK2 of the terminal **10** included in the stored CO for AP. Since the decryption of the digital signature DSpr succeeds in the case of FIG. 6, the AP **6** executes a second AP determination process that is regarding whether or not a specific value obtained by decrypting

the digital signature DSpr is identical to a value obtained by hashing the respective information in the received SCont for printer (that is, the hash value HV, the “Group”, and the public key PPK2). In the case of FIG. 6, the AP 6 determines “identical” in the second AP determination process, thus it determines that the authentication of the respective information in the DReq succeeds, and executes processes from T404 onward. The fact that the AP 6 determines “identical” in the second AP determination process means that the respective information in the received SCont for printer (that is, the hash value HV, the “Group”, and the public key PPK2) has not been tampered by a third party since the CO for printer was stored in the printer 100. On the other hand, in a case where the AP 6 determines “not identical” in the first AP determination process, in a case where the decryption of the digital signature DSpr fails, or in a case where the AP 6 determines “not identical” in the second AP determination process, the AP 6 determines that the authentication fails and does not execute the processes from T404.

(74) Next, in T404, the AP 6 creates a connection key CK (that is, a shared key) by using the obtained public key PPK2 of the printer 100 and the stored secret key ask2 of the AP 6 in accordance with the ECDH.

(75) In T410, the AP 6 sends a DPP Peer Discovery Response (hereinbelow termed simply as “DRes”) including the SCont for AP to the printer 100.

(76) In response to receiving the DRes from the AP 6 in T410 via the Wi-Fi I/F 116, the printer 100 executes a process for authenticating the sender of the DRes (that is, the AP 6) and the respective information in the DRes (that is, the hash value HV, the “Group1”, and the public key APK2). Specifically, in T412, the printer 100 firstly executes a first PR determination process that is regarding whether or not the hash value HV and the group ID “Group” in the received SCont for AP are respectively identical to the hash value HV and the group ID “Group” in the SCont for printer included in the stored CO for printer. In the case of FIG. 6, the printer 100 determines “identical” in the first PR determination process, thus it determines that the authentication of the sender of the DRes (that is, the AP 6) succeeds. The fact that the hash value HV in the received SCont for AP is identical to the hash value HV in the SCont for printer included in the stored CO for printer means that the SCont for printer and the SCont for AP were created by the same device (that is, the terminal 10). As such, the printer 100 also determines that authentication of the creator of the received SCont for AP (that is, the terminal 10) succeeds. Further, the printer 100 decrypts the digital signature DSap in the received SCont for AP by using the public key TPK2 of the terminal 10 included in the stored CO for printer. Since the decryption of the digital signature DSap succeeds in the case of FIG. 6, the printer 100 executes a second PR determination process that is regarding whether or not a specific value obtained by decrypting the digital signature DSap is identical to a value obtained by hashing the respective information in the received SCont for AP (that is, the hash value HV, the “Group”, and the public key APK2). In the case of FIG. 6, the printer 100 determines “identical” in the second PR determination process, thus it determines that the authentication of the respective information in the DRes succeeds, and executes processes from T414 onward. The fact that the printer 100 determines “identical” in the second PR determination process means that the respective information in the SCont for AP (that is, the hash value HV, the “Group”, and the public key APK2) has not been tampered by a third party since the CO for AP was stored in the AP 6. On the other hand, in a case where the printer 100 determines “not identical” in the first PR determination process, in a case where the decryption of the digital signature DSap fails, or in a case where the printer 100 determines “not identical” in the second PR determination process, the printer 100 determines that the authentication fails and does not execute the processes from T414.

(77) In T414, the printer 100 creates a connection key CK by using the stored secret key psk2 of the printer 100 and the public key APK2 of the AP 6 in the received SCont for AP in accordance with the ECDH. Here, the connection key CK created by the AP 6 in T404 and the connection key CK created by the printer 100 in T414 are identical to each other. Due to this, the connection key

CK for establishing the DPP connection is shared between the printer **100** and the AP **6**. When **T414** is completed, the process of **FIG. 6** is terminated.

(78) As above, after the connection key CK has been shared between the printer **100** and the AP **6**, in **T40** of **FIG. 2** the printer **100** and the AP **6** use the connection key CK to execute communication of a 4 way-handshake. As a result, a DPP connection is established between the printer **100** and the AP **6**. The printer **100** changes the WFD flag **138** from “ON” to “OFF” in a case of establishing the DPP connection with the AP **6**.

(79) (Case B; **FIG. 7**)

(80) Next, the process of Case B in which a WFD connection is established between the terminal **10** and the printer **100** will be described with reference to **FIG. 7**. **T500** to **T512** are similar to **T100** to **T112** of **FIG. 3**. In response to the “terminal” button being selected by the user in the selection screen in **T520**, the terminal **10** executes a search process for searching for a device that is a connection target of the terminal **10** (that is, the printer **100**) in **T522**. Specifically, the following processes are executed.

(81) Firstly, the terminal **10** sends a Probe Request (hereinbelow termed simply as “PReq”) by broadcast via the Wi-Fi I/F **16**. When the terminal **10** sends the PReq by broadcast, the terminal **10** receives a respective Probe Response (hereinbelow termed simply as “PRes”) from one or more devices including the printer **100**. In this case, the terminal **10** specifies the printer **100** that is the connection target by specifying a PRes including the SSID “wfd” received in **T510** from among the one or more PRes. Then, the terminal **10** sends a PReq including the SSID “wfd” of the specified printer **100** to the printer **100** via the Wi-Fi I/F **16** (that is, sends by unicast).

(82) In response to receiving the PReq including the SSID “wfd” from the terminal **10**, the printer **100** sends a PRes to the terminal **10** via the Wi-Fi I/F **116**.

(83) In **T530**, the terminal **10** executes various communications with the printer **100** (Provision Discovery, Association, WPS Negotiation, 4 way-handshake). In the WPS Negotiation, the terminal **10** receives wireless setting information including the SSID “wfd” and password being stored in the memory **134** from the printer **100**. Then, the terminal **10** executes the 4 way-handshake communication with the printer **100** by using the SSID “wfd” and the password to establish a WFD connection with the printer **100**.

(84) In a case where a WFD connection with the terminal **10** is established in **T530**, the printer **100** shifts from the respondent state to the non-respondent state in **T540**. In a situation where a WFD connection is established between the terminal **10** and the printer **100**, a possibility is low that communication according to the DPP scheme is executed between the terminal **10** and the printer **100**. In such a situation, since the printer **100** shifts from the respondent state to the non-respondent state, it is possible to suppress maintenance of the respondent state which has a higher processing load than the non-respondent state.

(85) In response to accepting a print operation at **T550** from the user for causing the printer **100** to execute printing, in **T552** the terminal **10** sends print data to the printer **100** via the Wi-Fi I/F **16** by using the established WFD connection.

(86) In a case of receiving the print data from the terminal **10** via the Wi-Fi I/F **116** in **T552**, the printer **100** causes the print executing unit **120** to execute printing according to the print data in **T554**. When the process of **T554** is completed, the process of **FIG. 7** is terminated.

(87) (Case C; **FIG. 8**)

(88) Next, a process of Case C will be described with reference to **FIG. 8**. In Case C, the terminal **10** and the printer **100** has established a normal Wi-Fi connection with the AP **6**. For this reason, the memory **34** of the terminal **10** and the memory **134** of the printer **100** are storing AP information for establishing the normal Wi-Fi connection with the AP **6**. The AP information includes an SSID “ap6” for identifying the wireless network formed by the AP **6**. In an initial state of **FIG. 8**, since the AP information is being stored in the memory **134**, the printer **100** is operating in the non-setting mode.

(89) **T602** and **T602** are similar to **T100** and **T102** of FIG. 3. In **T604**, the printer **100** acquires a signal indicating that the NFC connection is established from the NFC I/F **118**, determines that the operation mode of the printer **100** is the non-setting mode, and supplies the WFD information including the SSID “wfd” and the AP information including the SSID “ap6” to the NFC I/F **118**.

T606 is similar to **T106** of FIG. 3. In **T610**, the NFC I/F **118** of the printer **100** sends the WFD information and the AP information to the terminal **10** by using the established NFC connection.

(90) In **T610**, in a case of receiving the WFD information and the AP information from the printer **100** via the NFC I/F **118**, the terminal **10** determines that the SSID “ap6” included in the received AP information is identical to the SSID “ap6” in the AP information stored in the memory **34**. In this case, the terminal **10** determines that communication with the printer **100** is currently possible and, in **T612**, displays a notification screen indicating that communication with the printer **100** is currently possible.

(91) **T650** to **T654** are similar to **T550** to **T554** of FIG. 7. In **T660**, the printer **100** determines that a predetermined time has elapsed since the WFD **138** was changed to “ON” in **T606** and, in **T662**, changes the WFD **138** from “ON” to “OFF”. When the process of **T662** is completed, the process of FIG. 8 is terminated.

(92) As above, in a case where an NFC connection is established with the terminal **10** (**T602**) in a situation of a normal Wi-Fi connection with the AP **6** being established, the printer **100** supplies the AP information to the NFC I/F **118** (**T604**). Due to this, the AP information is sent from the NFC I/F **118** to the terminal **10**, so that the terminal **10** can use the received AP information to determine whether communication with the printer **100** is currently possible. Then, in a case of determining that communication with the printer **100** is currently possible, the terminal **10** displays the notification screen. That is, in a situation where the terminal **10** and the printer **100** are currently capable of communicating with each other, the printer **100** does not execute the process for establishing a DPP connection between the printer **100** and the AP **6**, or the process for establishing a WFD connection between the terminal **10** and the printer **100**. As such, the processing load of the terminal **10** and the printer **100** can be reduced.

(93) (Case D; FIG. 9)

(94) Next, the process of Case D, in which a mode shifting operation is executed in the printer **100**, will be described with reference to FIG. 9. In Case D, the printer **100** has established a normal Wi-Fi connection with the AP **6**. As such, the printer **100** is storing the AP information including the SSID “ap6” in the memory **134**. Further, in an initial state of FIG. 9, the printer **100** is operating in the non-setting mode.

(95) In a case where a shift instruction for shifting the operation mode of the printer **100** is accepted from the user in **T700**, the printer **100** shifts from the non-setting mode to the setting mode in **T702**. **T703** and **T704** are similar to **T103** and **T104** of FIG. 3.

(96) In **T705**, the printer **100** acquires a signal from the NFC I/F **118** indicating that the NFC connection has been established, determines that the operation mode of the printer **100** is the setting mode, and supplies the WFD information and the BS information to the NFC I/F **118**. **T706** to **T712** are similar to **T106** to **T112** of FIG. 3. That is, the WFD information and the BS information are sent from the NFC I/F **118** of the printer **100** to the terminal **10**, and a selection screen is displayed on the terminal **10**. When the process of **T712** is completed, the process of FIG. 9 is terminated.

(97) As above, in a situation where a normal Wi-Fi connection is established between the printer **100** and the AP **6**, that is, in a situation where the printer **100** is operating in the non-setting mode, the printer **100** shifts to the setting mode in a case where the shift instruction is accepted from the user (**T702**). Then, the printer **100** supplies the WFD information and the BS information to the NFC I/F **118** in response to the establishment of the NFC connection with the terminal **10** (**T705**). As a result, the WFD information and the BS information are sent from the NFC I/F **118** to the terminal **10** (**T710**). Due to this, in a situation where the user desires to execute communication

according to the DPP scheme between the terminal **10** and the printer **100**, for example, in a situation where the user desires to establish a DPP connection between the printer **100** and an AP different from the AP **6**, the user can cause communication according to the DPP scheme to be executed between the terminal **10** and the printer **100** by executing the shift instruction in the printer **100** that is operating in the non-setting mode.

(98) (Process of Terminal **10**; FIG. **10**)

(99) Next, a process executed by the CPU **32** of the terminal **10** in order to realize the processes of FIGS. **2** to **9** will be described with reference to FIG. **10**. A process of FIG. **10** is executed in a case where an NFC connection is established between the NFC I/F **18** of the terminal **10** and the NFC I/F **118** of the printer **100**.

(100) In **S10**, the terminal **10** determines whether the BS information has been received from the printer **100** via the NFC I/F **18** by using the established NFC connection. In a case where the WFD information and the BS information are received from the printer **100** (for example, **T110** of FIG. **3**), the terminal **10** determines YES in **S10**, and proceeds to **S15**. On the other hand, in a case where the WFD information and the AP information are received from the printer **100** (for example, **T610** of FIG. **8**), the terminal **10** determines NO in **S10**, and proceeds to **S35**.

(101) In **S15**, the terminal **10** displays the selection screen on the display unit **14** (for example, **T112** of FIG. **3**).

(102) In **S20**, the terminal **10** determines whether the “terminal” button in the selection screen has been selected by the user. In a case where the “terminal” button in the selection screen has been selected by the user (for example, **T520** of FIG. **7**), the terminal **10** determines YES in **S20**, and proceeds to **S25**. On the other hand, in a case where the “AP” button in the selection screen has been selected (for example, **T120** of FIG. **3**), the terminal **10** determines NO in **S20**, and proceeds to **S30**.

(103) In **S25**, the terminal **10** executes various communications with the printer **100** (Probe, Provision Discovery, Association, WPS Negotiation, 4 way-handshake) according to the WFD scheme via the Wi-Fi I/F **16**, and establishes a WFD connection with the printer **100** (for example, **T522** and **T530** of FIG. **7**). When the process of **S25** is completed, the process of FIG. **10** is terminated.

(104) In **S30**, the terminal **10** executes the Auth and the Config with the printer **100** via the Wi-Fi I/F **16** (FIG. **4** and FIG. **5**). When the process of **S30** is completed, the process of FIG. **10** is terminated.

(105) In **S35**, the terminal **10** determines whether communication with the printer **100** is currently possible by using the received AP information. In a case of determining that communication with the printer **100** is currently possible, the terminal **10** determines YES in **S35** and, in **S40**, causes the display unit **14** to display a notification screen (for example, **T612** of FIG. **8**). On the other hand, in a case of determining that communication with the printer **100** is not currently possible, the terminal **10** determines NO in **S35** and, in **S25**, establishes a WFD connection with the printer **100** by using the received WFD information. When the process of **S25** or **S40** is completed, the process of FIG. **10** is terminated.

(106) (Process of Printer **100**; FIG. **11**)

(107) Next, a process executed by the CPU **132** of the printer **100** in order to realize the processes of FIGS. **2** to **9** will be described with reference to FIG. **11**. The process of FIG. **11** is executed in a case where an NFC connection is established between the NFC I/F **18** of the terminal **10** and the NFC I/F **118** of the printer **100**.

(108) In **S100**, the printer **100** determines whether a normal Wi-Fi connection with the AP **6** is being established. In a case where the AP information is being stored in the memory **134**, the printer **100** determines YES in **S100**, and proceeds to **S103**. On the other hand, in a case where the AP information is not being stored in the memory **134**, the printer **100** determines NO in **S100**, and proceeds to **S135**.

(109) In **S103**, the printer **100** determines whether it is operating in the non-setting mode. In case of determining that the printer **100** is operating in the non-setting mode (YES in **S103**), the printer **100** proceeds to **S105**. On the other hand, in case of determining that the printer **100** is operating in the setting mode (NO in **S103**), the printer **100** proceeds to **S135**.

(110) In **S105**, the printer **100** supplies the WFD information and the AP information to the NFC I/F **118** (for example, **T604** of FIG. 8). The WFD information includes an SSID for identifying the wireless network in which the printer **100** that is operating in the G/O state operates as a parent station. The AP information includes an SSID for identifying the wireless network in which the AP with which the printer **100** has established a normal Wi-Fi connection operates as a parent station.

(111) In **S110**, the printer **100** determines whether the WFD flag **138** stored in the memory **134** indicates “ON”. In a case where the WFD flag **138** indicates “ON”, the printer **100** determines YES in **S110**, and proceeds to **S120**. On the other hand, in a case where the WFD flag **138** indicates “OFF”, the printer **100** determines NO in **S110**, and proceeds to **S115**.

(112) In **S115**, the printer **100** changes the WFD flag **138** stored in the memory **134** from “OFF” to “ON” (for example, **T606** of FIG. 8). As a result, the printer **100** operates in the device state.

(113) In **S120**, the printer **100** shifts from the device state to the G/O state (for example, **T606** of FIG. 8).

(114) In **S125**, the printer **100** monitors receipt of a PReq from the terminal **10** via the Wi-Fi I/F **116**. In a case of receiving a PReq from the terminal **10**, the printer **100** determines YES in **S125**, and proceeds to **S130**. On the other hand, in a case where a predetermined time elapses without receiving a PReq from the terminal **10** (for example, **T660** of FIG. 8), the printer **100** determines NO in **S125**, and proceeds to **S132**.

(115) In **S130**, the printer **100** executes various communications (Probe, Provision Discovery, Association, WPS Negotiation, 4 way-handshake) according to the WFD scheme with the terminal **10**, and establishes a WFD connection with the terminal **10**. When the process of **S130** is completed, the process of FIG. 11 is terminated.

(116) In **S132**, the printer **100** changes the WFD flag stored in the memory **134** from “ON” to “OFF” (for example, **T662** of FIG. 8). When the process of **S132** is completed, the process of FIG. 11 is terminated.

(117) In **S135**, the printer **100** supplies the WFD information and the BS information to the NFC I/F **118** (for example, **T105** of FIG. 3). The BS information includes the public key of the printer **100**, the channel list, and the MAC address of the printer **100**.

(118) **S140** and **S145** are similar to **S110** and **S115**. In **S150**, the printer **100** shifts from the device state to the G/O state, and shifts from the non-respondent state to the respondent state (for example, **T106** and **T108** of FIG. 3).

(119) **S155** is similar to **S125**. In case of receiving a PReq from the terminal **10** via the Wi-Fi I/F **116** (for example, **T522** of FIG. 7), the printer **100** determines YES in **S155**, and proceeds to **S160**. On the other hand, in case of receiving an AReq from the terminal **10** via the Wi-Fi I/F **116** (for example, **T210** of FIG. 4), the printer **100** determines NO in **S155**, and proceeds to **S170**.

(120) **S160** is similar to **S130**. In **S165**, the printer **100** shifts from the respondent state to the non-respondent state (for example, **T540** of FIG. 7). When the process of **S165** is completed, the process of FIG. 11 is terminated.

(121) In **S170**, the printer **100** executes the Auth and the Config with the terminal **10** (for example, FIG. 4 and FIG. 5), and executes the NA and 4 way-handshake with the AP **6** (for example, FIG. 6, **T40** of FIG. 2) to establish a DPP connection with the AP **6**.

(122) In **S175**, the printer **100** changes the WFD flag stored in the memory **134** from “ON” to “OFF”. When the process of **S175** is completed, the process of FIG. 11 is terminated.

Effect of the Present Embodiment

(123) According to the present embodiment, the printer **100** supplies the BS information including the public key PPK1 of the printer **100**, and the WFD information including the SSID “wfd” to the

NFC I/F **118** (T**105** of FIG. 3). As a result, in a case where an NFC connection with the printer **100** is established (T**104** of FIG. 3), the terminal **10** receives the BS information and the WFD information from the printer **100** (T**110**), and displays the selection screen (T**112**). In a case where the “AP” button in the selection screen is selected by the user (T**120**), that is, in a case of determining that a DPP connection is to be established between the printer **100** and the AP **6**, the terminal **10** sends an AReq to the printer **100** (T**210** of FIG. 4). In a case of receiving the AReq from the terminal **10**, the printer **100** sends an ARes to the terminal **10** (T**220**). In a case of receiving the ARes from the printer **100**, the terminal **10** sends a CRes including the CO for printer to the printer **100** (T**310** of FIG. 5). In a case of receiving the CRes from the terminal **10**, the printer **100** establishes a DPP connection between the printer **100** and the AP **6** by using the CO for printer (T**35** and T**40** of FIG. 2). That is, the printer **100** can establish a Wi-Fi connection according to the DPP scheme (that is, a DPP connection) with the AP **6**. On the other hand, in a case where the “terminal” button in the selection screen is selected by the user (T**520** of FIG. 7), that is, in a case of determining that a WFD connection is to be established between the terminal **10** and the printer **100**, the terminal **10** sends a PReq including the SSID “wfd” included in the WFD information to the printer **100** (T**522**). In a case of receiving the PReq from the terminal **10**, the printer **100** establishes a WFD connection between the terminal **10** and the printer **100** (T**530**). That is, the printer **100** can establish a WFD connection according to the WFD scheme with the terminal **10**. As such, in response to the establishment of an NFC connection between the printer **100** and the terminal **10**, the printer **100** can establish a DPP connection or a WFD connection different from the NFC connection according to an appropriate connection scheme among plurality of connection schemes.

Correspondence Relationship

(124) The printer **100**, the terminal **10**, the AP **6** are an example of “a communication device”, “a first external device”, “a second external device”, respectively. The NFC I/F **18**, the NFC I/F **118** are an example of “a first wireless interface” of the “first external device”, “a first wireless interface” of the “communication device” respectively, and the Wi-Fi I/F **16** is an example of “a second wireless interface” and “a third wireless interface” of the “first external device”. The Wi-Fi I/F **116** is an example of the “second wireless interface” and “third wireless interface” of the “communication device”. The public key PPK**1**, the SSID “wfd” of the printer **100** are an example of “a first public key (or a public key)”, “specific information”, respectively. The NFC connection of T**104** of FIG. 3, the DPP connection of T**40** of FIG. 2, the WFD connection of T**530** of FIG. 7 are an example of “a first wireless connection”, “a second wireless connection”, “a third wireless connection”, respectively. The DPP scheme, the WFD scheme are an example of “first connection scheme”, “second connection scheme”, respectively. The AReq of T**210**, the ARes of T**220**, the CO for printer of FIG. 4 are an example of “a first authentication request (or an authentication request)”, “a first authentication response (or an authentication response)”, “first connection information (or connection information)”, respectively. The PReq including the SSID “wfd” sent from the terminal **10** in the search process of T**522** is an example of the “specific information”.

(125) The print data of T**552** of FIG. 7 is an example of “object data”. The non-setting mode, the setting mode are an example of “a first mode”, “a second mode”, respectively. The AP information of T**610** of FIG. 8 is an example of “determination information”. The case where NO is determined in S**20** of FIG. 10 is an example of “a case where it is determined that the second wireless connection is to be established”, and the case where YES is determined in S**20** is an example of “a case where it is determined that the third wireless connection is to be established”.

(126) The process of T**504** of FIG. 7, the process of S**135** of FIG. 11, the process of T**220** of FIG. 4, the process of T**310** of FIG. 5, the process of T**35** and T**40** of FIG. 2, and the process of T**530** of FIG. 7 are an example of processes executed by “establish a first wireless connection”, “supply a first public key that is a public key of the communication device and specific information to the first wireless interface”, “send a first authentication response that is a response to the first

authentication request to the first external device via the second wireless interface”, “receive first connection information from the first external device via the second wireless interface”, “establish, according to a first connection scheme, the second wireless connection between the communication device and the second external device via the second wireless interface by using the first connection information”, “establish, according to a second connection scheme different from the first connection scheme, the third wireless connection between the communication device and the first external device via the third wireless interface” of the “communication device”, respectively. (127) The process of **T504** of FIG. 7, the process of **S10**, the process of **S20** of FIG. 10, the process of **T210**, the process of **T220** of FIG. 4, the process of **T310** of FIG. 5, the process of **T522**, the process of **T530** of FIG. 7 are an example of processes executed by “establish a first wireless connection”, “receive a public key of the communication device and specific information from the communication device by using the first wireless connection via the first wireless interface”, “determine which of the second wireless connection and the third wireless connection is to be established”, “send an authentication request in which the public key is used to the communication device via the second wireless interface”, “receive an authentication response that is a response to the authentication request from the communication device via the second wireless interface”, “send connection information to the communication device via the second wireless interface”, “send a specific signal including the specific information to the communication device via the third wireless interface”, “establish, according to a second connection scheme different from the first connection scheme, the third wireless connection via the third wireless interface between the communication device and the first external device” of the “first external device”, respectively.

Second Embodiment; FIG. 12 and FIG. 13

(128) Next, a second embodiment will be described with reference to FIGS. 12 and 13. The second embodiment differs from the first embodiment in that the terminal **10** executes the process of FIG. 12 instead of the process of FIG. 10, and the printer **100** executes the process of FIG. 12 instead of the process of FIG. 11. Firstly, the process executed by the CPU **32** of the terminal **10** will be described with reference to FIG. 12.

(129) **S210** is similar to **S10** of FIG. 10. In a case of receiving the WFD information and the BS information from the printer **100**, the terminal **10** determines YES in **S210**, and proceeds to **S215**. **S215** to **S230** are similar to **S15** to **S30** of FIG. 10. On the other hand, in a case of receiving a Read Command according to the NFC scheme from the printer **100**, the terminal **10** determines NO in **S210**, and proceeds to **S235**.

(130) In **S235**, the terminal **10** supplies, to the NFC I/F **18**, the public key of the terminal **10**, the channel list indicating the plurality of communication channels determined in advance in the terminal **10** (that is, the plurality of communication channels that can be used by the terminal **10**), and the MAC address of the terminal **10**. As a result, the NFC I/F **18** sends the public key, the channel list, and the MAC address to the printer **100** by using the established NFC connection.

(131) In **S240**, the terminal **10** changes a value of the capability of the terminal **10** from a value indicating that the terminal **10** is capable of operating as the Configurator to a value indicating that the terminal **10** is capable of operating as the Enrollee.

(132) In **S245**, the terminal **10** shifts from a non-respondent state to a respondent state. That is, the terminal **10** starts a process of monitoring receipt of an AReq using one communication channel among the plurality of communication channels in the channel list.

(133) In **S250**, firstly, the terminal **10** executes the Auth and the Config with the printer **100**. In the Auth, the terminal **10** receives the AReq from the printer **100**, sends an ARes to the printer **100**, receives the Confirm from the printer **100**, and determines to operate as the Enrollee. Further, in the Config, the terminal **10** sends a CReq to the printer **100**, receives a CRes including a CO for terminal from the printer **100**, and stores the CO for terminal for establishing a DPP connection between the terminal **10** and the AP **6**. Next, by using the CO for terminal, the terminal **10** executes the NA with the AP **6**, shares, with the AP **6**, a connection key for establishing a DPP connection

with the AP 6, executes a 4 way-handshake with the AP 6, and establishes a DPP connection with the AP 6. When the process of S250 is completed, the process of FIG. 12 is terminated.

(134) (Process of Printer; FIG. 13)

(135) Next, a process executed by the CPU 132 of the printer 100 will be described with reference to FIG. 13. S300 is similar to S100 of FIG. 11. The printer 100 proceeds to S305 in a case of determining that a normal Wi-Fi connection with the AP 6 is being established (YES in S300). On the other hand, the printer 100 proceeds to S335 in a case of determining that a normal Wi-Fi connection with the AP 6 is not being established (NO in S300). S335 to S375 are similar to S135 to S175 of FIG. 11.

(136) In S305, the printer 100 changes a value of the capability of the printer 100 from the value indicating that the printer 100 is capable of operating as the Enrollee to the value indicating that the printer 100 is capable of operating as the Configurator.

(137) In S310, the printer 100 causes the NFC I/F 118 to send a Read Command. As a result, the NFC I/F 118 sends the Read Command to the terminal 10 by using the established NFC connection.

(138) In S315, the printer 100 receives the public key, the channel list, and the MAC address of the terminal 10 from the terminal 10 via the NFC I/F 118.

(139) In S320, the printer 100 executes the Auth and the Config with the terminal 10. In the Auth, the printer 100 sends an AReq to the terminal 10, receives an ARes from the terminal 10, sends the Confirm to the terminal 10, and determines to operate as the Configurator. Further, in the Config, the printer 100 receives a CReq from the terminal 10, creates a CO for terminal, and sends a CRes including the CO for terminal to the terminal 10. When the process of S320 is completed, the process of FIG. 13 is terminated.

Effect of the Present Embodiment

(140) In this embodiment, in a case of determining that a normal Wi-Fi connection with the AP 6 is being established (YES in S300), the printer 100 sends a Read Command to the terminal 10 via the NFC I/F 118 (S310), receives the public key, the channel list, and the MAC address of the terminal 10 from the terminal 10 (S315), executes the Auth and the Config with the terminal 10 via the Wi-Fi I/F 116, and sends the CO for terminal to the terminal 10 (S320). As a result, the terminal 10 can receive the CO for terminal from the printer 100, execute an NA with the AP 6 by using the CO for terminal, and execute a 4 way-handshake with the AP 6 to establish a DPP connection with the AP 6. Due to this, the printer 100 becomes capable of executing communication with the terminal 10 via the AP 6.

Correspondence Relationship

(141) The public key of the terminal 10 of S315 of FIG. 13 is an example of “a second public key”. The AReq sent from the printer 100 to the terminal 10 and the ARes sent from the terminal 10 to the printer 100 in the Auth of S320 are an example of “a second authentication request”, “a second authentication response”, respectively. The CO for terminal is an example of “second connection information”.

(142) (Variant 1) In T35 of FIG. 2, the process of NA may be executed between the terminal 10 and the printer 100 to establish a DPP connection between the terminal 10 and the printer 100. That is, “a second external device” may be the same device as the “first external device”.

(143) (Variant 2) In the above embodiments, the terminal 10 is used to establish a DPP connection between the printer 100 and the AP 6. Alternatively, for example, the terminal 10 may be used to establish a DPP connection between the printer 100 and another device operating as the G/O (that is, a device operating as a parent station). Further, for example, the terminal 10 may be used to establish a DPP connection between the printer 100 operating as the G/O (that is, a device operating as a parent station) and another device (that is, a device operating as a child station). That is, the “second external device” may not be “an access point”.

(144) (Variant 3) The terminal 10 and the printer 100 may further comprise a BT I/F for executing a

wireless communication according to the Bluetooth (registered trademark, hereinbelow termed simply as “BT”) scheme. The above BT scheme includes BT scheme version 4.0 and above (so-called Blue Tooth Low Energy). In this case, for example, in T105 of FIG. 3, the printer 100 supplies, instead of the WFD information, BT information for establishing a wireless connection according to the BT scheme (hereinbelow termed simply as “BT connection”) to the NFC I/F 118. As a result, the NFC I/F 118 sends the BT information and the BS information to the terminal 10 by using the NFC connection. In a case of receiving the BT information and the BS information from the printer 100, the terminal 10 causes the display unit 14 to display a selection screen in response to the “terminal” button being selected in the selection screen, and establishes a BT connection with the printer 100 by using the received BT information. In the present variant, the BT I/F of the terminal 10 is an example of “a third wireless interface” of the “first external device”, and the BT I/F of the printer 100 is an example of “a third wireless interface” of the “communication device”. Further, the BT scheme, and the BT connection are an example of “a second connection scheme”, “a third wireless connection”, respectively.

(145) (Variant 4) The printer 100 may also operate in the setting mode in a case where a normal Wi-Fi connection with the AP 6 is established. In the present variant, “shift the operation mode of the communication device from a first mode to a second mode”, “supply the specific information and determination information to the first wireless interface” may be omitted.

(146) (Variant 5) The process of S165 of FIG. 11 (or S365 of FIG. 13) may be omitted. In the present variant, “shift the operation state of the communication device from the respondent state to the non-respondent state” may be omitted.

(147) (Variant 6) The process of S15 of FIG. 10 (or S215 of FIG. 12) may be omitted. In this case, in a case of receiving the WFD information and the BS information from the printer 100, the terminal 10 may determine whether the public key TPK2 and the secret key tsK2 are being stored in the memory 34, that is, may determine whether the Config with the AP 6 has been executed. In case of determining that the Config with the AP 6 has been executed, the terminal 10 executes the Auth and the Config with the printer 100 in S30. On the other hand, in case of determining that the Config with the AP 6 has not been executed, the terminal 10 establishes a WFD connection with the printer 100 in S25. In the present variant, the case of determining that the Config with the AP 6 has been executed is an example of “a case where it is determined that the second wireless connection is to be established”, and the case of determining that the Config with the AP 6 has not been executed is an example of “a case where it is determined that the third wireless connection is to be established”.

(148) (Variant 7) In S105 of FIG. 11, the printer 100 may supply the MAC address of the printer 100, instead of the AP information, to the NFC I/F 118. In a case of receiving the WFD information and the MAC address from the printer 100 via the NFC I/F 118, the terminal 10 determines that the BS information has not been received from the printer 100 (NO in S10 of FIG. 10) and, in S35, determines whether communication with the printer 100 is currently possible by using the received MAC address. Specifically, the terminal 10 firstly determines whether a normal Wi-Fi connection with the AP 6 has been established. In a case of determining that a normal Wi-Fi connection with the AP 6 has not been established, the terminal 10 determines NO in S35, and proceeds to S25. On the other hand, in a case of determining that a normal Wi-Fi connection with the AP 6 has been established, the terminal 10 sends a request signal by broadcast via the Wi-Fi I/F 16 by using the AP 6. In response to the sending of the request signal by broadcast, the terminal 10 receives respective response signals to the request signal from each of the one or more devices connected to the AP 6. In this case, the terminal 10 determines whether a response signal including the received MAC address of the printer 100 is included in the one or more response signals. In a case of determining that the response signal including the MAC address of the printer 100 is included in the one or more response signals, the terminal 10 determines YES in S35, and proceeds to S40. On the other hand, in a case of determining that the response signal including the MAC address of the

printer **100** is not included in the one or more response signals, the terminal **10** determines NO in S35, and proceeds to S25. In the present variant, the MAC address of the printer **100** is an example of the “determination information”.

(149) (Variant 8) In the above embodiments, the terminal **10** executed each process according to the app **38**, but may execute the following processes according to the OS program **36**. That is, the terminal **10** establishes an NFC connection with the printer **100**, and receives the WFD information and the BS information from the printer **100** via the NFC I/F **18** according to the OS program **36**. In this case, the terminal **10** determines whether the app **38** has been activated. In a case of determining that the app **38** has not been activated, the terminal **10** determines that a DPP connection is to be established between the printer **100** and the AP **6**, and executes the same processes as in FIG. 4 and FIG. 5. On the other hand, in a case of determining that the app **38** has been activated, the terminal **10** determines that a WFD connection is to be established between the terminal **10** and the AP **6**, and executes the same processes as in T522 and T530 of FIG. 7. In the present variant, also, in the case of receiving an AReq from the terminal **10**, the printer **100** can send an ARes to the terminal **10**, receive a CRes including the CO for printer from the terminal **10**, and establish a DPP connection with the AP **6** by using the CO for printer. On the other hand, in the case of receiving a PReq including the SSID “wfd” from the terminal **10**, the printer **100** can establish a WFD connection with the terminal **10**. As such, in response to the establishment of the NFC connection between the printer **100** and the terminal **10**, the printer **100** can establish a DPP connection or WFD connection different from the NFC connection according to an appropriate connection scheme among the plurality of connection schemes.

(150) (Variant 9) The process (for example, T202, T212 of FIG. 4) for creating the shared key (for example, SK1) is not limited to the process described above according to the ECDH, but may be another process according to the ECDH. Further, the process for creating the shared key is not limited to processes according to the ECDH, and a process according to another scheme (for example, DH (Diffie-Hellman key exchange), etc.) may be executed. Further, in the above embodiments, the digital signatures DSap and DSpr were created according to ECDSA, but may be created according to another scheme (for example, DSA (Digital Signature Algorithm), RAS (Rivest-Shamir-Adleman cryptosystem) etc.).

(151) (Variant 10) Instead of the NFC I/F **18**, the terminal **10** may comprise a different wireless interface according to a wireless scheme different from the NFC scheme (for example, BT scheme, TransferJet scheme). In this case, the printer **100** may also comprise, instead of the NFC I/F **118**, a different wireless interface according to the wireless scheme different from the NFC scheme. For example, in T110 of FIG. 3 the printer **100** sends the WFD information and the BS information to the terminal **10** via that wireless interface. In the present variant, the different wireless interface according to the wireless scheme different from the NFC scheme is an example of the “first wireless interface”.

(152) (Variant 11) For example, instead of operating as the G/O of the WFD scheme, the printer **100** may operate as a so-called Soft AP. In this case, for example, in T105 of FIG. 3, the printer **100** creates an SSID and password used in the wireless network in which the printer **100** operates as the Soft AP, and supplies Soft AP information including the SSID and password to the NFC I/F **118**. As a result, in T110, the NFC I/F **118** sends the Soft AP information and the BS information to the NFC I/F **18** by using the established NFC connection. In the present variant, the SSID in the Soft AP information is an example of the “specific information”.

(153) (Variant 12) In the case of determining NO in S155 of FIG. 11 (or S355 of FIG. 13), that is, in the case of receiving an AReq from the terminal **10** via the Wi-Fi I/F **116** (for example, T210 of FIG. 4), the printer **100** may change the WFD flag from “ON” to “OFF” before executing the process of S170. In this case, when the process of S170 (or S370) ends, the printer **100** terminates the process of FIG. 11 (or FIG. 13). In a situation where an AReq is received from the terminal **10**, the possibility is low of a communication according to the WFD scheme being executed between

the terminal **10** and the printer **100**. In such a situation, since the printer **100** changes the WFD flag from “ON” to “OFF”, it is possible to suppress a state in which the WFD flag indicates “ON”, that is, to suppress maintenance of a state in which the printer **100** can operate according to the WFD scheme.

(154) (Variant 13) The “communication device” may not be a printer, but may be another device such a scanner, a multi-function peripheral, a portable terminal, a PC, a server, etc.

(155) (Variant 14) In each of the above embodiments, the respective processes of FIGS. **2** to **13** are implemented by software (that is, the app **38**, the program **136**). Instead of this, one or more of the processes may be implemented by hardware such as a logic circuit.

Claims

1. A communication device comprising: one or more wireless interfaces; a processor; and a memory storing computer-readable instructions therein, the computer-readable instructions, when executed by the processor, causing the communication device to: accept a predetermined operation; in a case where the predetermined operation is accepted, shift an operation state of the communication device from a non-respondent state to a respondent state and from a non-capable state to a capable state, the non-respondent state being a state of not sending an authentication response to a first external device even when an authentication request is received from the first external device, the respondent state being a state of sending the authentication response to the first external device in response to receiving the authentication request from the first external device, the non-capable state being a state incapable of operating according to a second connection scheme different from a first connection scheme, and the capable state being a state capable of operating according to the second connection scheme; in a case where the operation state of the communication device is the respondent state and the authentication request is received from the first external device via a first wireless interface among the one or more wireless interfaces: send the authentication response to the first external device via the first wireless interface; receive first connection information from the first external device via the first wireless interface after the authentication response has been sent to the first external device, the first connection information being for establishing a first wireless connection between the communication device and a second external device via the first wireless interface; and in a case where the first connection information is received from the first external device, establish, according to the first connection scheme, the first wireless connection between the communication device and the second external device via the first wireless interface by using the first connection information; and in a case where the operation state of the communication device is the capable state and a specific signal based on specific information is received from the first external device via a second wireless interface among the one or more wireless interfaces, shift to a state capable of communicating with the first external device via the second wireless interface according to the second connection scheme different from the first connection scheme.

2. The communication device as in claim 1, wherein the predetermined operation is an operation for establishing a second wireless connection according to a third connection scheme between the communication device and the first external device.

3. The communication device as in claim 1, wherein the first connection scheme is a connection scheme according to a Device Provisioning Protocol (DPP) scheme.

4. The communication device as in claim 1, further comprising an output unit, wherein the computer-readable instructions, when executed by the processor, further cause the communication device to: cause the output unit to output a public key that is a public key of the communication device and the specific information, wherein the public key and the specific information are acquired by a first external device due to the public key and the specific information being outputted, the public key is used in the authentication request, and the specific signal is based on

the specific information.

5. The communication device as in claim 4, wherein after the operation state of the communication device has been shifted from the non-respondent state to the respondent state and from the non-capable state to the capable state, the output unit is caused to output the public key and the specific information.

6. The communication device as in claim 1, wherein the computer-readable instructions, when executed by the processor, further cause the communication device to: after the operation state of the communication device has been shifted from the non-respondent state to the respondent state and from the non-capable state to the capable state, operate as a parent station of the second connection scheme.
